Rearrangement of a formula

1 Make x the subject of the following equations:

a $y = x + z$ **b** $y = \frac{x}{-6}$ **c** $y = 7(2 + x)$ **d** $\frac{x}{2} + \frac{n}{5} = 2$

e $3y = 2x + z$ **f** $y = \frac{-x}{7}$ **g** $y = 5(3 - x)$ **h** $\frac{x}{5} - \frac{m}{7} = 2$

2 Make m the subject of the following equations:

a $\frac{m}{y} = gh$ **b** $y = 4mx - 9$ **c** $x = 5k(n + m)$ **d** $\frac{2m}{p} = st$

e $b = -2am + 10$ **f** $x = -7m(p + q)$ **g** $y = -5mx - 14$ **h** $x = -7k(l - m)$

3 Make y the subject of the following equations:

a $\frac{x}{y} = h - k$ **b** $9x^2 + 3y = 12$

c $\frac{3a}{b} = 2j - y$ **d** $4x^2 - 2y = -16$

4 Make k the subject of the following equations:

a $m = \frac{9kx}{8} - 6$ **b** $r = \frac{k}{k - 9}$

c $v = \frac{5kw}{7} - 3$ **d** $s = \frac{2k}{k + 4}$

5 Make R the subject of $V = IR - E$.

6 Make r the subject of $L = \frac{E}{R + r}$.

7 Make p the subject of the equation $M = pq - 5pr^2$.

8 Consider the Pythagorean formula $c^2 = a^2 + b^2$.

a Make a^2 the subject of the formula. **b** Make b^2 the subject of the formula.

9 Make r^2 the subject of the following formulas:

a The volume of a cone is given by the formula $V = \frac{1}{3}\pi r^2 h$.

b The area of a circle is given by the formula $A = \pi r^2$.

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c The area of a sector of a circle is given by the formula $A = \dfrac{\theta \pi r^2}{360}$.

10 Make v^2 the subject of the formula for kinetic energy, $K = \dfrac{1}{2}mv^2$.

11 The bend allowance for sheet metal is given by the formula $B = 2\pi\left(R + \dfrac{T}{2}\right) \times \dfrac{A}{360}$.

Make T, the thickness of the sheet, the subject of the formula.

Applications

12 Temperature conversion from degrees Fahrenheit to degrees Celsius is given by the formula $C = \dfrac{5}{9}(F - 32)$.

The temperature inside a freezer is $5\,^{\circ}\mathrm{F}$. Find this temperature in degrees Celsius.

13 The perimeter of a rectangle is given by the formula $P = 2(a + b)$.

Find the value of b if $a = 5$ and $P = 26$.

14 The surface area of a rectangular prism is given by the formula $S = 2(ab + bh + ah)$.

Find the value of h if $a = 3$, $b = 3$ and $S = 138$.

15 The surface area of a rectangular prism is given by the formula $S = 2(lw + wh + lh)$, where l, w and h are the dimensions of the prism.

Find the surface area of a rectangular prism with a length of $8\ \mathrm{cm}$, a width of $9\ \mathrm{cm}$ and a height of $5\ \mathrm{cm}$.

16 The surface area of a cylinder is given by the formula $S = 2\pi r^2 + 2\pi rh$.

Find the surface area of a cylinder with a diameter of $10\ \mathrm{cm}$ and a height of $25\ \mathrm{cm}$.

17 The volume of a sphere is given by the formula $V = \dfrac{4}{3}\pi r^3$, where r is the radius of the sphere. Find the radius of a sphere with a volume of $124\ \mathrm{mm}^3$ to two decimal places.

18 Newton's second equation of motion is $s = ut + \dfrac{1}{2}at^2$.

Find the value of a if $s = 2596$, $u = 250$ and $t = 22$.

19 The power of a circuit is given by the formula $P = I^2R$.

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Find the value of I, correct to two decimal places, when $P = 40$ and $R = 560$.

20 The resistance of a parallel circuit is given by the formula $\frac{1}{a} = \frac{1}{b} + \frac{1}{c}$.

Find the value of b, correct to two decimal places, if $a = 8$ and $c = 23$.

21 The reactance of a capacitor is given by the formula: $X = \frac{1}{2\pi fC}$.

Find the value of f, correct to one decimal place, if $X = 44.1$ and $C = 190 \times 10^{-9}$.

22 The gravitational force between two masses is given by $F = \frac{GMm}{d^2}$.

Find the value of M, correct to two decimal places, if $F = 15$, $G = 6.67 \times 10^{-11}$, $m = 1$ and $d = 5 \times 10^{-6}$.

23 Amelia bought a television series online. The television series was 4.8 gigabytes large and took 4 hours to download.

a Find the size of the television series in gigabits. Use the formula $b = 8B$, where b is the number of gigabits and B is the number of gigabytes.

b Hence, find the average number of gigabits downloaded each hour.

24 The life expectancy, y, of a person born in Japan is approximated by the equation $y = 0.27x + 72$, where x is the number of years since 1970.

In what year will the life expectancy in Japan reach 95 years?

25 The amount of oil, y, imported to the United States from Canada in millions of barrels per day can be approximated by the equation $y = 0.105x + 1.34$, where x is the number of years since 2000.

In what year will the approximate number of barrels imported from Canada be 5.5 million per day?

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